

Concurrent and Distributed Systems

Lab 1 - Java Threads

Tasks

1. Read the slides **Processes and Threads.pptx** and try the example programs in the slides (you can find them in **cds.lab1.zip**).
2. Get familiar with Java threads, following the examples or “recipes” from **Java 9 Concurrency Cookbook** (see below).
 - Each example is introduced by explaining step-by-step how to write the Java classes, having the pieces of code provided.
 - There are also GitHub repositories / publisher website providing the complete Java classes for each book chapter example. However, the best way to learn is by building yourselves these programs while reading the explanations from the book.
 - For each example you have the key ingredients in the book, plus useful **Notes** and **Tips** to help learning more about the Java concurrency framework.
 - If you remain stuck at an example, please ask for help.

Java Concurrency Cookbook - Chapter 1. Thread Management

During the lab session it might be difficult to cover all the examples from Chapter 1. You can start with those highlighted in **bold and with [*]**, and continue later with other examples.

- **Creating, running, and setting the characteristics of a thread [*]**
- **Interrupting a thread [*]**
- Controlling the interruption of a thread
- **Sleeping and resuming a thread [*]**
- Waiting for the finalization of a thread
- Creating and running a daemon thread
- **Processing uncontrolled exceptions in a thread [*]**
- **Using local thread variables [*]**
- Grouping threads and processing uncontrolled exceptions in a group of threads
- Creating threads through a factory

Solutions

The solutions are in **cds.lab1.zip**. The archive contains **the examples from the lecture slides and all the recipes from Java 9 Concurrency Cookbook - Chapter 1** (with minimal refactoring of the source code from the repository).

Resources

- J. Fernandez Gonzalez (2017), **Java 9 Concurrency Cookbook** (2nd edition), Packt Publishing. You can access the online version through the library (log in with your university account), link [here](#). The GitHub repository providing the complete Java classes for each book chapter example is here: <https://github.com/PacktPublishing/Java-9-Concurrency-Cookbook-Second-Edition>

- J. Friesen (2015), **Java threads and concurrency utilities**, Apress. You can access the online version through the library (log in with your university account), link [here](#).
- B. Wittman, T. Korb, A. Mathur (2013), **Start Concurrent: An Introduction to Problem Solving in Java with a Focus on Concurrency**, Purdue University Press. Book also available in HTML format here <https://start-concurrent.github.io/>. You can jump directly to [Chapter 13. Concurrent Programming](#) or to any other Java topics you want to revise.